



University of
Nottingham

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COMP-2024 / COMP-2039

Artificial Intelligence Methods

Lecture 2b

Reading: Performance Analysis of
Stochastic Local Search Methods



Learning Outcomes

IDENTIFY

1. Main components of heuristic search/optimisation methods
2. Representation
3. Neighbourhoods
4. Evaluation Function
5. Hill climbing methods
6. **Reading: Performance Analysis of Stochastic Local Search Methods – Preliminaries**





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READING

Performance Analysis of Stochastic Local Search Methods – Preliminaries



Which Stochastic Search Algorithm Performs Better for Solving Problem X?

- Algorithm A is new, B & C are previous approaches applied to the instance Inst1 of the minimising problem X.
- Assume all algorithms are run for the same number of objective function evaluations, and experiments are fair.
- Each experiment is repeated for 30 times, that is, an algorithm is run for 30 times, independently, on an instance.

<u>Algorithm A</u>					<u>Algorithm B</u>				<u>Algorithm...</u>			
<u>Instance avg. std. med. min.</u>					<u>avg. std. med. min.</u>				avg...			
run/trial	A	B	C									
1	1	8	20	So, which algorithm performs the best for solving Problem X?	Inst1	0.9	0.7	1	0	5.7	3.3	6
2	1	6	11									
3	0	2	15									
4	1	1	12									
5	0	3	11									
6	2	4	13									
7	0	8	3									
8	1	6	19									
9	1	1	15									
10	0	8	3									
11	1	8	11									
12	2	11	9									
13	1	3	14									
14	2	11	12									
15	1	5	6									
• avg.: mean objective value computed by averaging the objective values of 30 independent trials/runs. • std.: standard deviation associated with avg. • med.: median objective value. • min.: objective value of the best solution found in all trials/runs.												

So, which algorithm performs the best for solving Problem X?

- avg.:** mean objective value computed by averaging the objective values of 30 solutions returned by an algorithm from 30 independent trials/runs.
- std.:** standard deviation associated with avg.
- med.:** median objective value.
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Which Stochastic Search Algorithm Performs Better for Solving Problem X?

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<u>Algorithm A</u>					<u>Algorithm B</u>				<u>Algorithm...</u>
<u>Instance avg. std. med. min.</u>					<u>avg. std. med. min.</u>				avg...
run/trial	A	B	C						
1	1	8	20						
2	1	6	11						
3	0	2	15						
4	1	1	12						
5	0	3	11						
6	2	4	13						
7	0	8	3						
8	1	6	19						
9	1	1	15						
10	0	8	3						
11	1	8	11						
12	2	11	9						
13	1	3	14						
14	2	11	12						
15	1	5	6						

Inst1	0.9	0.7	1	0	5.7	3.3	6	
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So, which algorithm performs the best for solving Problem X?

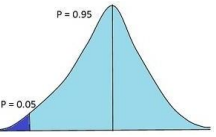
- Any comment for 1 instance is valid for 1 problem instance. So, Algorithm A performs the best for the instance with the label Inst1 of the problem X based on mean, median, best (minimum) objective values.



Which Stochastic Search Algorithm Performs Better for Solving Problem X?

- Apply **non-parametric statistical test – one tailed:**

- E.g.: Given two algorithms; X versus Y, $>$ ($<$) denotes that X (Y) is better than Y (X) and this performance difference is statistically significant within a confidence interval of 95% and $X \geq Y$ ($X \leq Y$) indicates that X (Y) performs better on average than Y (X) but no statistical significance (Wilcoxon signed rank test – e.g.,



<http://vassarstats.net/wilcoxon.html>

Stronger conclusions can be drawn for individual instances (like Inst1) when statistical tests validate the performance differences.

	<u>Algorithm A</u>						<u>Algorithm B</u>					<u>Algorithm...</u>
run/trial	A	B	C									
1	1	8	20									
2	1	6	11									
3	0	2	15									
4	1	1	12									
5	0	3	11									
6	2	4	13									
7	0	8	3									
8	1	6	19									
9	1	1	15									
10	0	8	3									
11	1	8	11									
12	2	11	9									
13	1	3	14									
14	2	11	12									
15	1	5	6									

Instance

avg.

std.

med.

min.

>

avg.

std.

med.

min.

avg...

Inst1

0.9

0.7

1

0

5.7

3.3

6

• A stronger conclusion can be provided for one instance

Important: Always repeat the experiments *more than or equal to*

for a meaningful statistical comparison.

Inst1 0.9 0.7 1 0 **>** 5.7 3.3 6 avg...

- A stronger conclusion can be provided for one instance (Inst1)**

Important: Always repeat the experiments *more than or equal to 30 times* for any given instance for a meaningful statistical comparison.



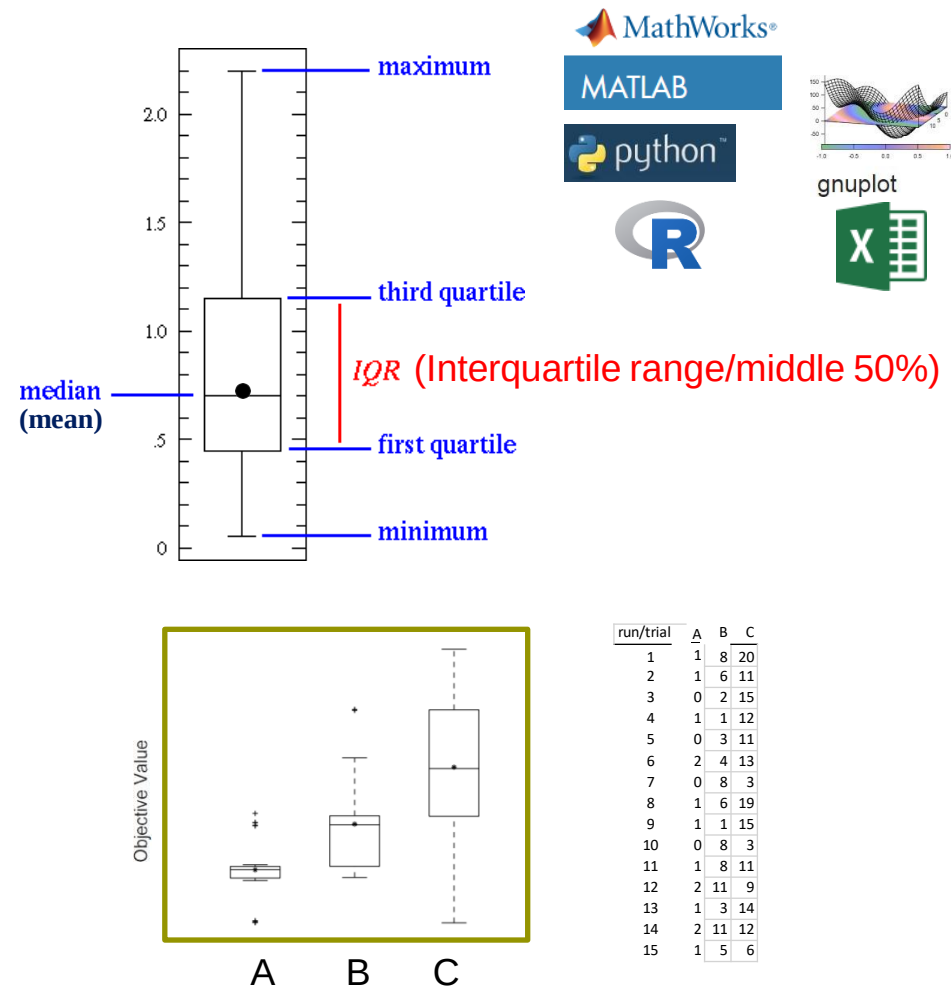
Which Stochastic Search Algorithm Performs Better for Solving Problem X? Boxplots

- **Boxplots** illustrates groups of numerical data through their quartiles.

BoxPlotR: <http://shiny.chemgrid.org/boxplotr/>

- Example: boxplot of objective values obtained from 30 runs on the instance Inst1 from 3 algorithms.

- Outliers may be plotted as individual points.
- **A stronger conclusion for Inst1 – next slide**





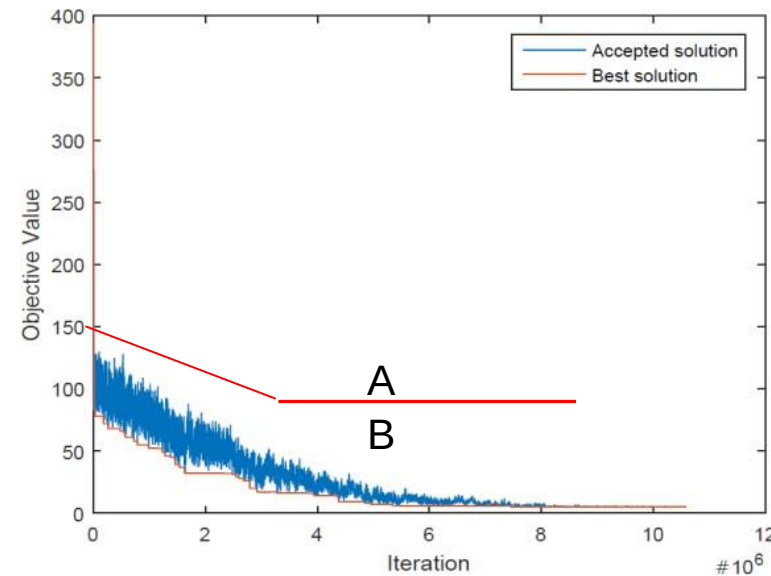
References – Box Plot

- <https://www.atlassian.com/data/charts/box-plot-complete-guide>
- <https://datatab.net/tutorial/box-plot>
- <https://www.geeksforgeeks.org/box-plot/>



Performance Analysis Using Plots – Other Methods

- **Progress plot (line plot)** – per instance: Objective value from a run or mean of objective values from multiple runs per iteration/time unit.



Objective value of best and accepted solutions versus iteration from Algorithms A and B for the same instance from a sample run.



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Performance Comparison of Stochastic Search Algorithms – Part 2



Which Stochastic Search Algorithm Performs Better for Solving Problem X?

run/trial	A	B	C
1	1	8	20
2	1	6	11
3	0	2	15
4	1	1	12
5	0	3	11
6	2	4	13
7	0	8	3
8	1	6	19
9	1	1	15
10	0	8	3
11	1	8	11
12	2	11	9
13	1	3	14
14	2	11	12
15	1	5	6
	⋮		
	⋮		
	⋮		
	⋮		

- Algorithm A is new, B & C are previous approaches applied to the instance Inst1 of the minimising problem X
- Assume all algorithms are run for the same number of objective function evaluations, and experiments are fair
- Each experiment is repeated for 30 times, that is, an algorithm is run for 30 times, independently, on an instance

Algorithm A					Algorithm B				Algorithm C			
Instance	avg.	std.	med.	min.	avg.	std.	med.	min.	avg.	std.	med.	min.
Inst1	0.9	0.7	1	0	5.7	3.3	6	1	11.6	4.95	12	3

Which algorithm performs best?

- avg.:** mean objective value computed by averaging the objective values of 30 solutions returned by an algorithm from 30 independent trials/runs
- std.:** standard deviation associated with avg.
- med.:** median objective value
- min.:** objective value of the best solution found in all trials/runs



Which Stochastic Search Algorithm Performs Better for Solving Problem X?

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1	1	8	20
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	⋮		

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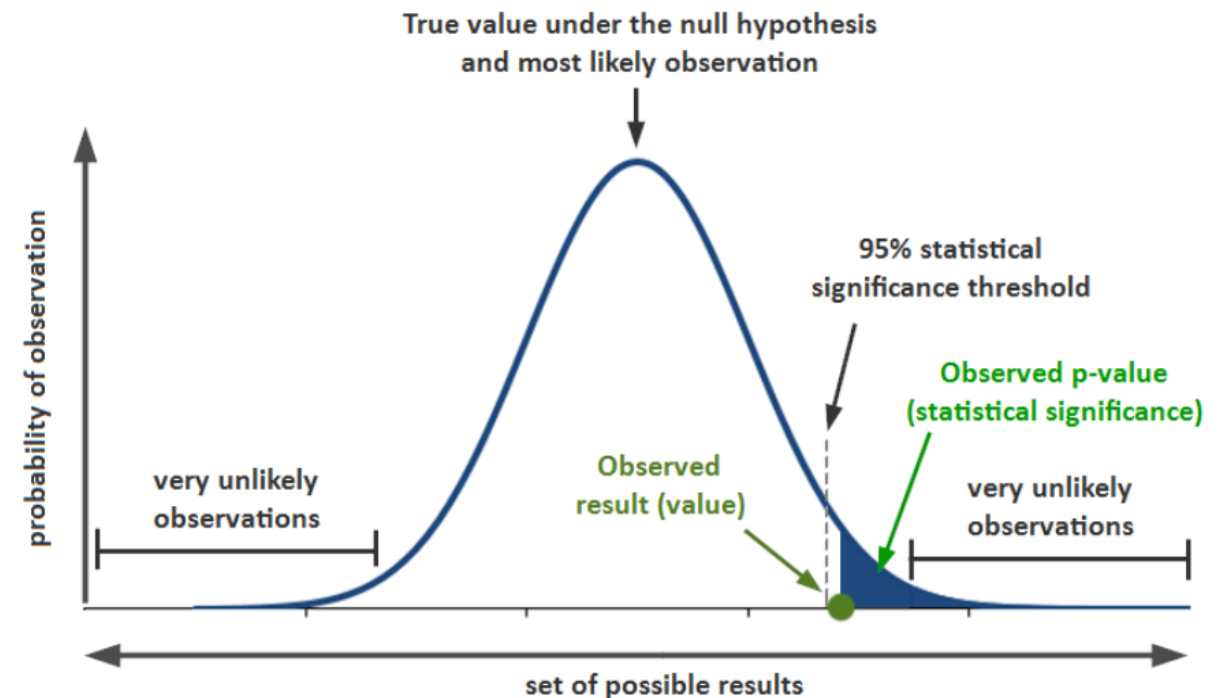
Which algorithm performs best?

Any comment for 1 instance is valid for 1 problem instance. So, **Algorithm A** performs the best for the instance with the label Inst1 of the problem X based on mean, median, best (minimum) objective values.



Statistical test, null hypothesis and p-values

- The null hypothesis states the results are due to chance and are not significant in terms of supporting the idea being investigated.
- A *p-value*, or probability value, is a number describing how likely it is that your data would have occurred by random chance (i.e., that the null hypothesis is true).



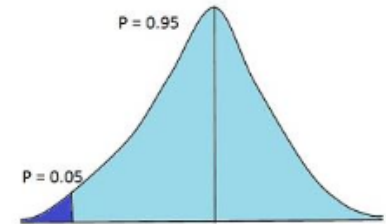


Which Stochastic Search Algorithm Performs Better for Solving Problem X?

run/trial	A	B	C
1	1	8	20
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12	2	11	9
13	1	3	14
14	2	11	12
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	⋮		

Apply non-parametric statistical test – one tailed:

- E.g.,: Given two algorithms; X versus Y, $>$ ($<$) denotes that X (Y) is better than Y (X) and this performance difference is statistically significant within a confidence interval of 95% and $X \geq Y$ ($X \leq Y$) indicates that X (Y) performs better on average than Y (X) but no statistical significance (Wilcoxon signed rank test – e.g., <http://vassarstats.net/wilcoxon.html>)



Instance	Algorithm A				vs.	Algorithm B				vs.	Algorithm C			
	avg.	std.	med.	min.		avg.	std.	med.	min.		avg.	std.	med.	min.
Inst1	0.9	0.7	1	0	>	5.7	3.3	6	1	>	11.6	4.95	12	3

- A stronger conclusion can be provided for one instance (Inst1)
- **Important:** Always repeat the experiments more than or equal to 30 times for any given instance for a meaningful statistical comparison.



Statistical Tests

Nonparametric statistical tests

Type of comparison	Procedures
Pairwise comparisons	Sign test Wilcoxon test
Multiple comparisons ($1 \times N$)	Multiple sign test Friedman test Friedman Aligned ranks Quade test Contrast Estimation
Multiple comparisons ($N \times N$)	Friedman test

Nonparametric statistics **are not based on any assumptions**, such as the sample following a specific distribution, and measures the central tendency with the median value

Post-hoc procedures

Type of comparison	Procedures
Multiple comparisons ($1 \times N$)	Bonferroni
	Holm
	Hochberg
	Hommel
	Holland
	Rom
	Finner
Multiple comparisons ($N \times N$)	Li
	Nemenyi
	Holm
	Shaffer
	Bergmann

J. Derrac, S. García, D. Molina, F. Herrera, A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms, Swarm and Evolutionary Computation, vol. 1, issue 1, pp. 3-18, 2011.

<https://www.semanticscholar.org/paper/A-practical-tutorial-on-the-use-of-nonparametric-as-Derrac-Garc%C3%ADa/eeeea4322184e54f04f79263eda451cb4272b8ee4>



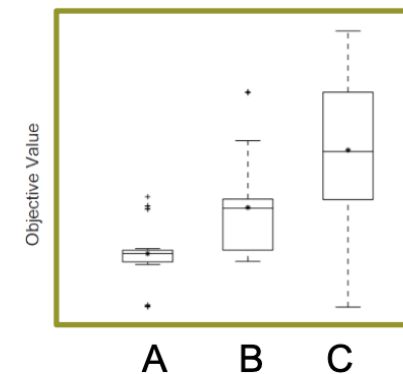
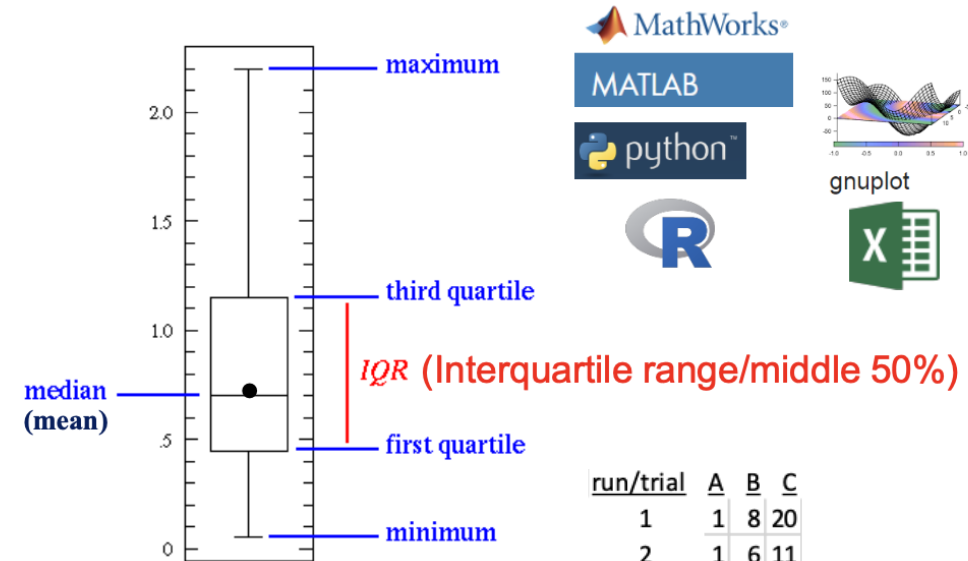
Statistical Tests

Aspect	Post-hoc Procedures	Non-Parametric Tests
When Applied	After finding significance in an overall test (e.g., ANOVA, Kruskal-Wallis).	When testing for significance initially.
Assumptions	Based on the assumptions of the overall test (e.g., parametric or non-parametric).	No assumption of normal distribution.
Purpose	To identify which groups are different while controlling for Type I error.	To determine whether differences exist between groups.
Example Use Case	After ANOVA/Kruskal-Wallis finds significance, perform pairwise comparisons using Nemenyi.	Use Kruskal-Wallis if ANOVA assumptions are violated.
Data Type	Follows the type of test (parametric or non-parametric).	Ordinal, ranked, or skewed interval data.



Which Stochastic Search Algorithm Performs Better for Solving Problem X? **Boxplots**

- Boxplots illustrates groups of numerical data via their quartiles. Example: **BoxPlotR** (<http://shiny.chemgrid.org/boxplotr/>)
- Example: boxplot of objective values obtained from 30 runs on the instance Inst1 from 3 algorithms
 - Outliers may be plotted as individual points
 - A stronger conclusion for Inst1 – next slide



run/trial	A	B	C
1	1	8	20
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8	1	6	19
9	1	1	15
10	0	8	3
11	1	8	11
12	2	11	9
13	1	3	14
14	2	11	12
15	1	5	6
...			

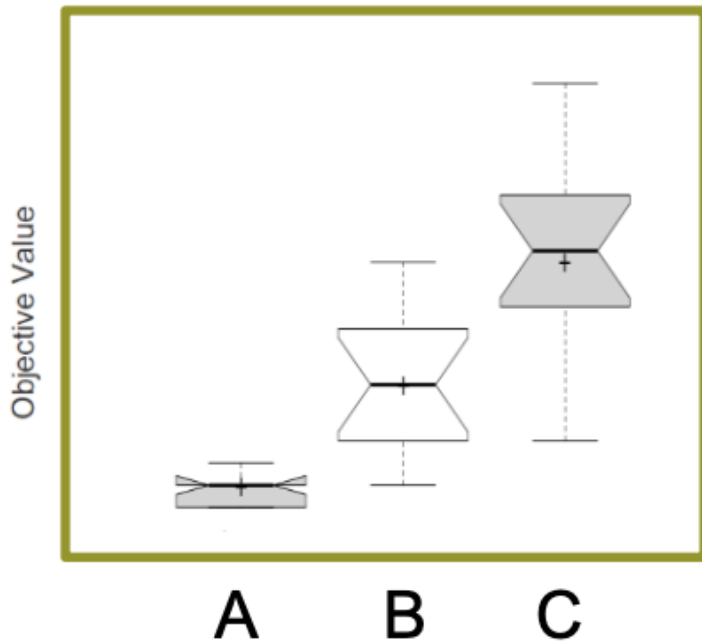


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- <https://www.geeksforgeeks.org/box-plot/>



Which Stochastic Search Algorithm Performs Better for Solving Problem X? **Notched Boxplots**

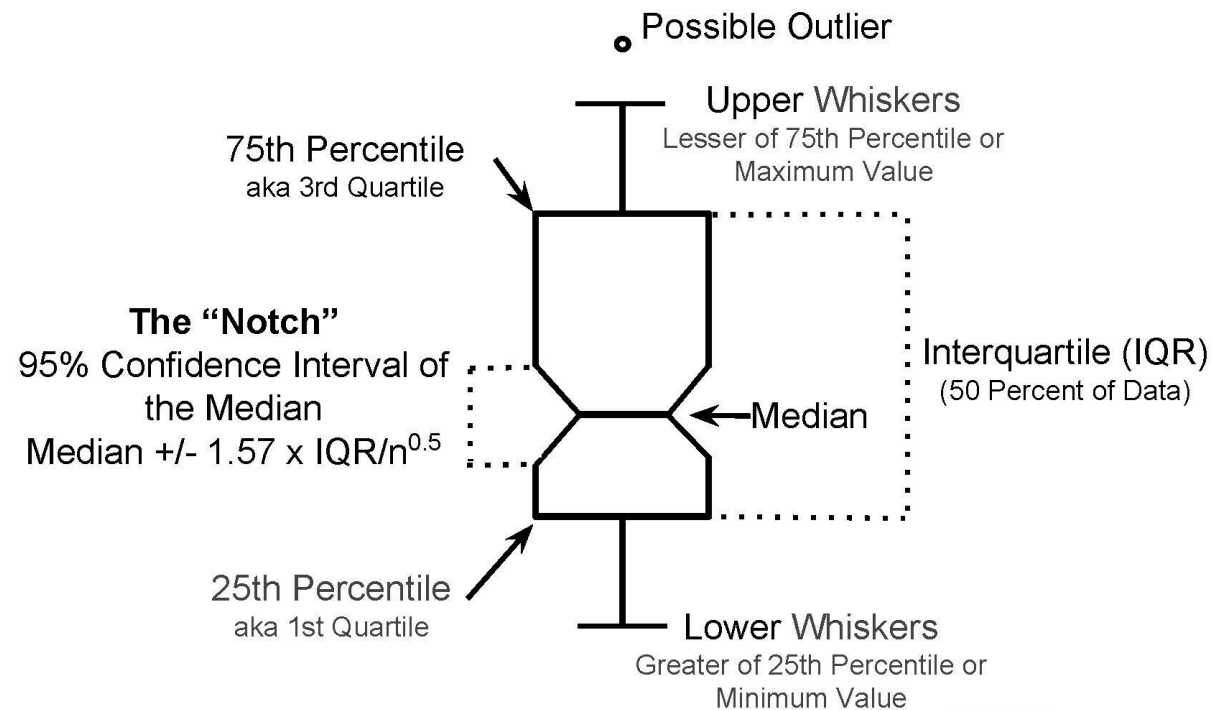


- The **notched boxplot** allows you to evaluate confidence intervals (by default 95% confidence interval) for the *medians* of each boxplot.
- Since the notches in the boxplots A vs B, A vs C and B vs C do not overlap, you can conclude that with 95% confidence, that the true medians do differ between each pair of those algorithms on Inst1: A performs significantly better than B as well as C, and B performs significantly better than C.



Notched boxplot

- <https://datavizcatalogue.com/blog/chart-snapshot-notched-box-plots/>
- <https://www.psychtc.org/psychtc/glossary2/notched-box-plot/>





Which Stochastic Search Algorithm Performs Better for Solving Problem X?

Algorithms are run on multiple instances from Problem X – now even a stronger conclusions can be made, e.g.,:

- Algorithm A performs the best on (this benchmark/ dataset) all problem X instances based on the mean/median objective values achieved over 30 trials.
- Algorithm C is the worst performing approach based on all metrics.
- Algorithm B provides the same best solution for 3 instances as Algorithm A.

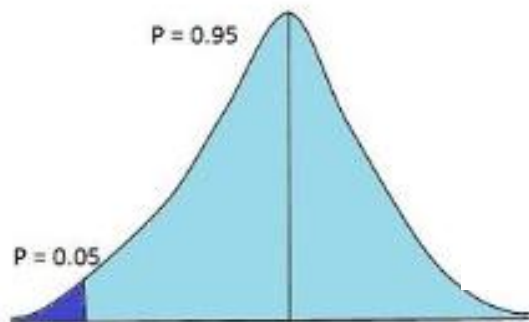
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Inst1	0.9	0.7	1	0	5.7	3.3	6	1	11.6	4.95	12	3
Inst2	3.1	3.9	2	1	21.3	13	12	3	44.9	9.8	30	18
Inst3	0.7	0.5	1	0	7.1	7.7	3	0	26.3	14	13	1
Inst4	1.7	1	1	1	5.7	4.3	3	1	20	4.6	15	12
Inst5	7.6	0.9	7	7	10.4	1.5	8	7	15.4	1.7	14	13



Which Stochastic Search Algorithm Performs Better for Solving Problem X?

Apply non-parametric statistical test – one tailed:

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Which Stochastic Search Algorithm Performs Better for Solving Problem X?

Algorithms are run on multiple instances from Problem X – now even a stronger conclusions can be made, e.g.,:

- Algorithm A is better than Algorithm B, while Algorithm B is better than C on (this benchmark/dataset or) all problem X instances on average and all those performance differences between the pair of algorithms are statistically significant within a confidence interval of 95%.

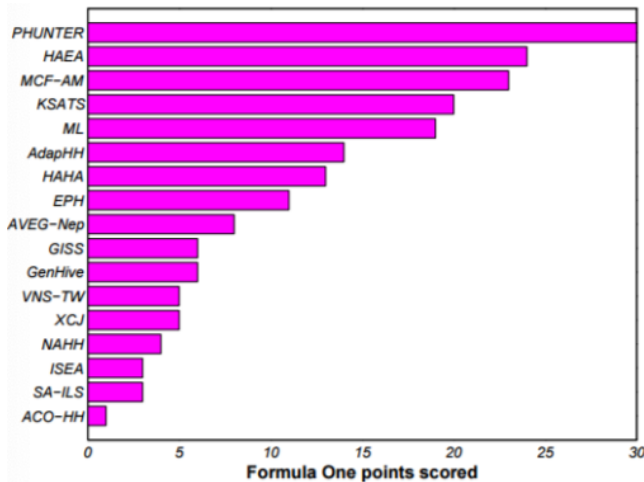
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Inst5	7.6	0.9	7	7	10.4	1.5	8	7	15.4	1.7	14	13



Performance Analysis Using Plots – Other Methods

- **Progress Plot** – per instance: *Objective value from a run or mean of objective values from multiple runs per iteration/time unit*
- **Ranking**

$$f_{norm}(s) = \frac{f(s) - f(s_{best})}{f(s_{worst}) - f(s_{best})}$$

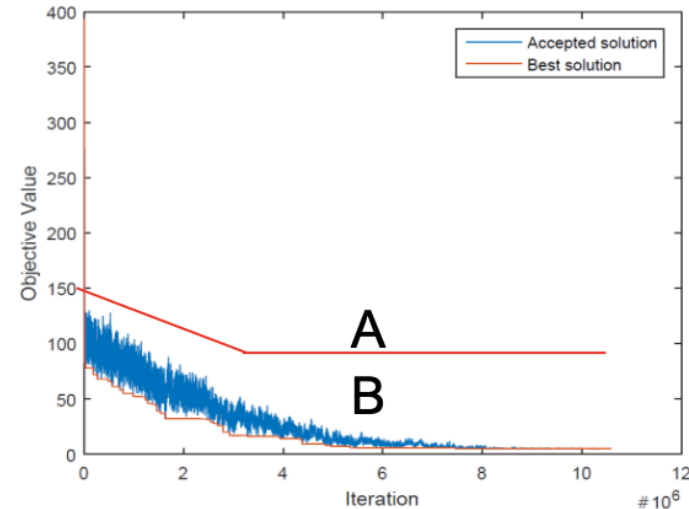


(higher score is better)

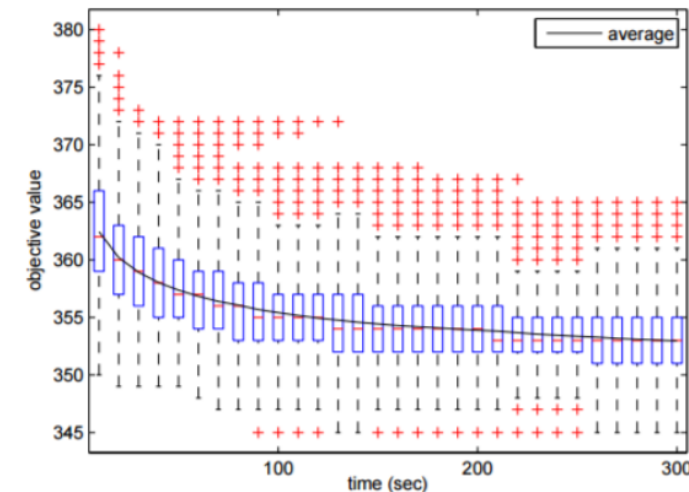
	Cross-domain
IE	17.67
AILLA	15.91
TA	20.04
GD	13.15
AILTA	19.05
NA	29.61
SA	10.10
SARH	10.22

(lower score is better)

Formula 1 vs normalised (mean) objective value ranking of algorithms on multiple instances



Objective value of best and accepted solutions versus iteration from Algorithms A and B for the same instance from a sample run



Box plot of objective values collected at each time step (every 10 sec.) from multiple runs of the Algorithm A on an instance



Summary



1. Main components of heuristic search/optimisation methods
2. Representation
3. Neighbourhoods
4. Evaluation Function
5. Hill climbing methods
6. Reading: Performance Analysis of Stochastic Local Search Methods – Preliminaries





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Questions



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NEXT:

Metaheuristics